The derived geometric Satake via factorization

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<u>Task 1</u> ~~$k = \bar{k}$~~, char $k = 0$.

G = reductive group $/k$, X = smooth curve $/k$

$x \in X(k)$

The non-derived geometric Satake equiv is a monoidal equiv. of abelian cats

$$\mathrm{Sat}_G^{\heartsuit} : \underset{\text{alg reps}}{\mathrm{Rep}(\check{G})^{\heartsuit}} \xrightarrow{\sim} \underset{\text{alg. D-modules}}{D(\mathcal{H}_{G,x})^{\heartsuit}}$$

where $\mathcal{H}_{G,x} = L^+G \backslash LG / L^+G$

$= \{ (\mathcal{P}_G, \mathcal{P}'_G, \alpha) : \mathcal{P}_G, \mathcal{P}'_G = G\text{-bundles on } D_x = \hat{X}_x$

and $\alpha : \mathcal{P}_G|_{\mathring{D}_x} \xrightarrow{\sim} \mathcal{P}'_G|_{\mathring{D}_x} \}$

is the Hecke stack at x.

$\mathrm{Rep}(\check{G})^{\heartsuit} \subset \mathrm{Rep}(\check{G})$ = DG cat of complexes w/ $\check{G}$-action.

$D(\mathcal{H}_{G,x})^{\heartsuit} \subset D(\mathcal{H}_{G,x}) =: \mathrm{Sph}_{G,x}$ = equivariant DG-cat. $D(\mathrm{Gr}_{G,x})^{L^+G}$.

Observation: $\mathrm{Rep}(\check{G})$ is the derived cat. of $\mathrm{Rep}(\check{G})^{\heartsuit}$. Thus $\mathrm{Sat}_G^{\heartsuit}$ extends uniquely to a t-exact monoidal functor

$$\mathrm{Sat}_G^{\text{naive}} : \mathrm{Rep}(\check{G}) \longrightarrow \mathrm{Sph}_{G,x}$$

This functor is <u>not</u> an equivalence.

For example, we have

$$\mathrm{REnd}_{\mathrm{Sph}_{G,x}}(\delta_1) = H^\bullet_{L^+G}(pt) = H^\bullet_G(pt) \simeq \mathrm{Sym}(\mathfrak{g}^*[-2])^G \quad \text{(Chern-Weil)}$$

whereas $\mathrm{REnd}_{\mathrm{Rep}(\check{G})}(k) = k$.

"Dericed Satake" means modifying $\mathrm{Rep}(\check{G})$ in order to upgrade $\mathrm{Sat}_G^{\mathrm{naive}}$ to an equiv.

<u>Theorem (Bezrukavnikov-Finkelberg)</u> There is a canonical monoidal equiv.

$$\begin{tikzcd} \mathrm{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}(\mathrm{Rep}(\check{G})) \arrow[r, "\sim"] & \mathrm{Sph}_{G,x}. \\ \mathrm{Rep}(\check{G}) \arrow[u, "\mathrm{ind}"] \arrow[ur, "\mathrm{Sat}_G^{\mathrm{naive}}"'] & \end{tikzcd}$$

Note that $\mathrm{REnd}_{\mathrm{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}(\mathrm{Rep}(\check{G}))}(\mathrm{Sym}(\check{\mathfrak{g}}[-2]))$

$$\simeq \mathrm{Sym}(\check{\mathfrak{g}}[-2])^{\check{G}} \simeq \mathrm{Sym}(\check{\mathfrak{t}}[-2])^W \simeq \mathrm{Sym}(\mathfrak{t}^*[-2])^W$$

$$\simeq \mathrm{Sym}(\mathfrak{g}^*[-2])^G.$$

<u>NB</u>: $\mathrm{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}(\mathrm{Rep}(\check{G}))$ is sym. mon, but $\mathrm{Sph}_{G,x}$ is <u>not</u> in a nat'l way. Namely, the assignment $x \mapsto \mathrm{Sph}_{G,x}$ upgrades to the structure of a factorization category.

I = nonempty finite set, $x_I = (x_i)_{i\in I} \in X^I$.

$$x_I \longmapsto \mathrm{Sph}_{G,x_I} := D(\mathcal{H}_{G,x_I})$$

$\varphi : I \twoheadrightarrow J \quad \rightsquigarrow \quad \Delta_\varphi : X^J \to X^I$,

$$\mathrm{Sph}_{G, x_J} \simeq \mathrm{Sph}_{G, \Delta_\varphi(x_J)}$$

That is, Sph_G naturally fibres over the Ran space $\mathrm{Ran} = \operatorname{colim}_I X^I$ parameterizing non empty finite subsets of X.

The factorization structure consists of a homotopy-coherent collection of equivalences

$$\mathrm{Sph}_{G, x_I} \simeq \bigotimes_{i \in I} \mathrm{Sph}_{G, x_i} \quad \text{whenever the } x_i \text{ are pairwise distinct.}$$

(Lurie tensor product)

Factorization categories assemble into a sym. mon. (∞-) category by applying Lurie tensor product fibrewise over Ran.

In particular, we can speak of monoidal factorization cat's = assoc. algebras in factorization categories.

Key example: Sph_G with its convolution product.

An aside: in the Betti setting, factorization categories (on $X = \mathbb{A}^1$, satisfying some condition) are equiv. to $\mathbb{E}_2$-monoidal cats. i.e. braided monoidal cats in the derived sense. A monoidal fact. cat. has an additional commuting $\mathbb{E}_1$-monoidal str. making it an $\mathbb{E}_1 + \mathbb{E}_2 = \mathbb{E}_3$-monoidal cat.

This explains (heuristically) why $\mathrm{Sph}_{G, x}$ is sym. mon: an $\mathbb{E}_3$-algebra in (1,1)-categories is already symmetric.

A standard construction attaches a fact. cat. to any sym. mon. DG cat. One can show that Sph_G does <u>not</u> arise in this way.

Goal: construct a monoidal fact. cat. $\mathrm{Sph}^{\mathrm{spec}}_{\check{G}}$ such that

$$\mathrm{Sph}^{\mathrm{spec}}_{\check{G},x} \simeq \mathrm{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}(\mathrm{Rep}(\check{G})) \quad \text{for } x \in X(k), \text{ and}$$

extend the B-F equiv. to the factorization setting.

Following Arinkin–Gaitsgory, we can define $\mathrm{Sph}^{\mathrm{spec}}_{\check{G},x}$ to be the "true" category of Hecke operators at x. Recall the global geom. Langlands conjecture:

$$\underset{\text{renormalized}}{D(\mathrm{Bun}_G(X))} \simeq \mathrm{IndCoh}(\underset{\substack{\text{moduli of flat } \check{G}\text{-bundles on } X \\ \underline{\text{derived}}}}{LS_{\check{G}}(X)}) \qquad (\text{here } X \text{ is assumed projective})$$

$$\mathcal{H}_{G,x} \curvearrowright \mathrm{Bun}_G(X)$$

Likewise, the spectral Hecke stack

$$\mathcal{H}^{\mathrm{spec}}_{\check{G},x} := \left\{ (E_{\check{G}}, E'_{\check{G}}, \alpha) : \begin{array}{l} E_{\check{G}}, E'_{\check{G}} = \text{flat } \check{G}\text{-bundles on } D_x, \\ \alpha : E_{\check{G}}|_{\mathring{D}_x} \xrightarrow{\sim} E'_{\check{G}}|_{\mathring{D}_x} \end{array} \right\}$$

acts on $LS_{\check{G}}(X)$

Thus it seems reasonable to put $\mathrm{Sph}^{\mathrm{spec}}_{\check{G},x} := \mathrm{IndCoh}(\mathcal{H}^{\mathrm{spec}}_{\check{G},x})$, and indeed we have $\mathrm{IndCoh}(\mathcal{H}^{\mathrm{spec}}_{\check{G},x}) \simeq \mathrm{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}(\mathrm{Rep}(\check{G}))$.

Namely, a choice of coord. at x determine an isom. of groupoids

$$\mathcal{H}^{\mathrm{spec}}_{\check{G},x} \simeq LS_{\check{G}}(D) \underset{LS_{\check{G}}(\mathring{D})}{\times} LS_{\check{G}}(D) \simeq \{0\}/\check{G} \underset{\check{\mathfrak{g}}/\check{G}}{\times} \{0\}/\check{G} \simeq (\{0\} \underset{\check{\mathfrak{g}}}{\times} \{0\})/\check{G}$$

Now $\{0\} \underset{\check{\mathfrak{g}}}{\times} \{0\} = \operatorname{Spec} \operatorname{Sym}(\check{\mathfrak{g}}^*[1])$, and

$\operatorname{IndCoh}(\{0\} \underset{\check{\mathfrak{g}}}{\times} \{0\})$ is compactly generated by the skyscraper / unit object δ_0, corresponding to the augmentation k.

Thus, $\operatorname{IndCoh}(\{0\} \underset{\check{\mathfrak{g}}}{\times} \{0\}) \simeq \operatorname{REnd}(\delta_0)\text{-mod} \simeq \operatorname{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}$.

Turning $\check{G}$-equiv. back on, we obtain the desired equivalence.

The stack $\mathcal{H}^{\operatorname{Spec}}_{\check{G},x}$ factorizes: a natural candidate for the fact. cat. would be $\operatorname{Sph}^{\operatorname{Spec}}_{\check{G},x^I} = \operatorname{IndCoh}(\mathcal{H}^{\operatorname{Spec}}_{\check{G},x^I})$.

Problem: $\mathcal{H}^{\operatorname{Spec}}_{\check{G},x^I}$ is not locally of finite type!

So defining coherent sheaves (as opposed to quasi-coherent) is problematic.

A toy model: $A = k[x, y, t, u, \frac{t-u}{x-y}, \frac{t-u}{(x-y)^2}, \cdots]$

$$\begin{tikzcd}
\mathbb{A}^1 \times \mathbb{A}^1 \arrow[r] \arrow[d] & \operatorname{Spec} A \arrow[d] & \mathbb{A}^2 \times (\mathbb{A}^2 \setminus \Delta) \arrow[l] \arrow[d] \\
\mathbb{A}^1 \arrow[r, "\Delta"] & \mathbb{A}^2 = \operatorname{Spec} k[x,y] & \mathbb{A}^2 \setminus \Delta \arrow[l]
\end{tikzcd}$$

<u>Talk 2</u>. Recall from last time: we want to extend $\operatorname{Sph}^{\operatorname{Spec}}_{\check{G},x} = \operatorname{IndCoh}(\mathcal{H}^{\operatorname{Spec}}_{\check{G},x}) \simeq \operatorname{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}\,(\operatorname{Rep}(\check{G}))$ to a factorization category, and to lift B-F to an equiv. of monoidal fact. categories $\operatorname{Sph}_G \simeq \operatorname{Sph}^{\operatorname{Spec}}_{\check{G}}$.

$(\operatorname{Sph}_G := D(\mathcal{H}_G)^{\operatorname{ren}})$

Heuristic: $f: \mathcal{Y} \to \mathcal{Z}$ affine morphism

$$\mathrm{QCoh}\left(\mathcal{Y}(\mathring{D}_{dR}) \underset{\mathcal{Z}(\mathring{D}_{dR})}{\times} \mathcal{Z}(D_{dR})\right) \approx f_* \mathcal{O}_{\mathcal{Y}}\text{-mod}^{\text{fact}}\left(\mathrm{QCoh}(\mathcal{Z}(D_{dR}))\right)$$

<u>Th. (Raskin)</u> For reasonable $\mathcal{Z}$ (eg. classifying stacks), this is literally true at a point.

Aside (Betti setting) A = comm ring

$$A\text{-mod}^{\mathbb{E}_n} \xrightarrow{\text{oblv}} \mathrm{Vect}$$

admits a left adjoint

<u>Prop.</u> The monad $\mathrm{Vect} \underset{\text{oblv}}{\overset{\dashrightarrow}{\leftarrow}} A\text{-mod}^{\mathbb{E}_n}$

is given by the assoc. alg $S^{n-1} \otimes A$.

$\Rightarrow$ $A\text{-mod}^{\mathbb{E}_n} \simeq S^{n-1} \otimes A\text{-mod}^{\text{left}}$

Note that $\mathrm{Spec}(S^{n-1} \otimes A) = \underline{\mathrm{Map}}(S^{n-1}, \mathrm{Spec}\, A)$

For $n=2$, we get a Betti version of the heuristic:

$$A\text{-mod}^{\mathbb{E}_2} \simeq \mathrm{QCoh}(\underline{\mathrm{Map}}(S^1, \mathrm{Spec}\, A))$$

Our case: $\mathcal{Y} = pt/G$, $\mathcal{Z} = pt/G \times pt/G$, $f = \Delta$.

$$\mathcal{Y}(\mathring{D}_{dR}) \underset{\mathcal{Z}(\mathring{D}_{dR})}{\times} \mathcal{Z}(D_{dR}) = LS_G(\mathring{D}) \underset{LS_G(\mathring{D}) \times LS_G(\mathring{D})}{\times} \left(LS_G(D) \times LS_G(D)\right)$$

$$= LS_G(D) \underset{LS_G(\mathring{D})}{\times} LS_G(D) = \mathcal{H}_G^{\mathrm{Spec}}$$

$\Rightarrow \mathrm{QCoh}(\mathcal{H}^{\mathrm{Spec}}_{G,x}) \simeq \mathcal{O}_G\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(G\times G))_x$

I = nonempty finite set

$\rightsquigarrow \mathcal{O}_G\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(G\times G))_{X^I}$

- $J = I \sqcup J_0$. $x_J = \{x_j\} \in X^J \rightsquigarrow M_{x_J} \in (\mathrm{Rep}(G\times G))$.

Comp as J varies, w/ iso's $M_{x_J} \simeq M_{x_I} \otimes \bigotimes_{j\in J_0} \mathcal{O}_G$

when $x_i \neq x_k$ for $j \neq k \in J_0$, and $x_i \neq x_j$ for $i \in I$, $j \in J_0$.

Renormalization

Why renormalize?

- to get IndCoh (as opposed to QCoh) at a point, needed for B-F
- to get a better category, eg. compactly generated
- to get a "ULA generated" category: once the derived Satake functor is constructed, can check that it is an equivalence at a point.

$\mathcal{C}$ = DG cat. w/ t-structure.

Standing assumptions:

- $\mathcal{C}^{\geq 0}$ stable under filtered colimits.
- (right completeness) $\operatorname{colim}_n \tau^{\leq n} C \xrightarrow{\sim} C$ for any $C \in \mathcal{C}$.

An object $C \in \mathcal{C}$ is called <u>almost compact</u> if $\tau^{\geq n} C$ is compact in $\mathcal{C}^{\geq n}$ for all n.

Call C <u>coherent</u> if it is almost compact and eventually coconnective.

(almost compact $\Rightarrow$ eventually connective by right completeness)

<u>Eg.</u> $\mathcal{C} = \mathrm{QCoh}(Y)$, Y = almost finite type scheme, $\mathcal{F} \in \mathrm{QCoh}(Y)$ is coherent in this sense $\Leftrightarrow$ coherent in usual sense.

We say that the t-structure on $\mathcal{C}$ is <u>coherent</u> if (in addition to standing assumptions)

- $\mathcal{C}^{\geq 0}$ is compactly generated;
- $\mathcal{C}^{\geq 1} \hookrightarrow \mathcal{C}^{\geq 0}$ preserves compact objects

  ($\Leftrightarrow$ coherent objects preserved by truncation)

Remark: $\mathcal{C}$ coherent $\Rightarrow$ $(\mathcal{C}^{\heartsuit})^c \subset \mathcal{C}^{\heartsuit}$ is abelian.

Definitions are from appendix to "Integral transforms for coherent sheaves" by Ben-Zvi, Nadler and Preygel.

If $\mathcal{C}$ is coherent and is compactly generated by $\mathcal{C}^{coh}$ = coherent objects, we call $\mathcal{C}$ <u>regular</u>.

NB: coherence is a property of $\mathcal{C}^+$ (not so for regularity).

If $\mathcal{C}$ is coherent, then it admits a canonical renormalization

$$\underset{\text{regular}}{\mathcal{C}^{ren}} := \mathrm{Ind}(\mathcal{C}^{coh}) \xrightarrow{\text{t-exact}} \mathcal{C}. \quad \text{such that } (\mathcal{C}^{ren})^+ \xrightarrow{\sim} \mathcal{C}^+.$$

Eg. $\mathcal{C} = \mathrm{QCoh}(Y)$, Y almost finite type. $\mathcal{C}$ coherent and $\mathrm{QCoh}(Y)^{ren} = \mathrm{IndCoh}(Y)$.

Talk 3. $\mathrm{Rep}(G)$ sym. mon. $\rightsquigarrow$ $\mathrm{Rep}(G)_{X^I}$ factorization cat.

has a unique t-structure s.t.

$\mathrm{Rep}(G)_{X^I} \longrightarrow D(X^I)$ is t-exact.

Prop. (Raskin) This t-structure is regular. (in particular, coherent).

Likewise, the cat. $\mathcal{O}_G\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(G\times G))_{X^I}$ has a unique t-structure s.t. forgetful functor to $\mathrm{Rep}(G\times G)_{X^I}$ is t-exact.

Th. (C.-Raskin) This t-structure is coherent, and the vacuum module $\mathcal{O}_{G,X^I}$ is "almost ULA".

S = smooth variety.

$D(S) \curvearrowright \mathcal{C}$

$$\mathrm{QCoh}(S) \underset{D(S)}{\otimes} \mathcal{C} \underset{\mathrm{oblv}_S}{\overset{\mathrm{ind}_S}{\rightleftarrows}} \mathcal{C}$$

We call $c \in \mathcal{C}$ ULA if $\mathrm{oblv}_S(c)$ is compact.

If $\mathrm{QCoh}(S) \underset{D(S)}{\otimes} \mathcal{C}$ is compactly generated by objects $\mathcal{P} \otimes \mathrm{oblv}_S(c_i)$ where $\mathcal{P}$ is perfect complex on S, we call $\mathcal{C}$ ULA generated.

Example. $\mathcal{C} = \mathrm{Rep}(G)_{X^I}$

$$\mathrm{Loc}_{X^I} : \mathrm{Rep}(G)^{\otimes I} \longrightarrow \mathrm{Rep}(G)_{X^I}$$

The objects $\mathrm{Loc}_{X^I}(\boxtimes_i V_i)$, V_i fin. dim. are ULA generators.

Suppose $\mathcal{C}$ has a t-structure comp. w/ the $D(S)$-action

(i.e. $\mathcal{C} \longrightarrow D(S)^\vee \otimes \mathcal{C} \simeq D(S) \otimes \mathcal{C}$ is t-exact). Then

$\mathrm{QCoh}(S) \underset{D(S)}{\otimes} \mathcal{C}$ has a unique t-structure s.t.

$$\mathrm{QCoh}(S) \underset{D(S)}{\otimes} \mathcal{C} \underset{\mathrm{oblv}_S}{\overset{\mathrm{Ind}_S}{\rightleftarrows}} \mathcal{C} \qquad \text{is t-exact}$$

Call $c \in \mathcal{C}$ <u>almost ULA</u> if $\mathrm{oblv}_S(c)$ is almost compact.

If $\left(\mathrm{QCoh}(S) \underset{D(S)}{\otimes} \mathcal{C}\right)^{\geq 0}$ is compactly generated by objects

$\tau^{\geq 0}(p \otimes \mathrm{oblv}_S(c_i))$ w/ c_i almost ULA, we call $\mathcal{C}$ <u>almost ULA gen'd</u>.

For $\mathcal{C}$ coherent and almost ULA gen.,

$D(S) \curvearrowright \mathcal{C}^{\mathrm{ren}}$ and $\mathcal{C}^{\mathrm{ren}}$ is ULA gen.

$$\begin{array}{ccc} \mathrm{Rep}(G)_{\times I} & \simeq & \mathcal{O}_G\text{-mod}\left(\mathrm{Rep}(G \times G)\right)_{\times I} \\ & & \Big\downarrow \mathrm{oblv}^{\mathrm{coh} \to \mathrm{fact}} \qquad \text{t-exact} \\ & & \mathcal{O}_G\text{-mod}^{\mathrm{fact}}\left(\mathrm{Rep}(G \times G)\right)_{\times I} \end{array}$$

<u>Th</u>. $\Rightarrow$ admits a $D(X^I)$-linear, monadic right adjoint (on bounded below objects).

$\Rightarrow \mathcal{O}_G\text{-mod}^{\mathrm{fact}}\left(\mathrm{Rep}(G \times G)\right)_{\times I}$ almost ULA gen.

$\Rightarrow \mathrm{Sph}^{\mathrm{Spec}}_{\check{G}, X^I} := \left(\mathcal{O}_{\check{G}}\text{-mod}^{\mathrm{fact}}\left(\mathrm{Rep}(\check{G} \times \check{G})\right)_{\times I}\right)^{\mathrm{ren}}$.

Over a point $x \in X(k)$:

$$\mathrm{Sph}^{\mathrm{Spec}}_{\check{G},x} = \mathrm{IndCoh}\left(\mathcal{H}^{\mathrm{Spec}}_{\check{G},x}\right) = \mathrm{IndCoh}\left(\{0\} \times_{\check{\mathfrak{g}}} \{0\}\right)^{\check{G}}$$

$$\mathrm{oblv}^{\mathrm{com}\to\mathrm{fact}} = \Delta_x : \mathrm{IndCoh}(pt/\check{G}) \to \mathrm{IndCoh}\left(\left(\{0\} \times_{\check{\mathfrak{g}}} \{0\}\right)/\check{G}\right)$$

<u>Rmks</u> i) $\mathrm{oblv}^{\mathrm{com}\to\mathrm{fact}} : \left(\mathrm{Rep}(G)_{XI}\right)^{\heartsuit} \xrightarrow{\sim} \left(\mathrm{Sph}^{\mathrm{Spec}}_{G,XI}\right)^{\heartsuit}$.

ii) monadicity of right adjoint of $\mathrm{oblv}^{\mathrm{com}\to\mathrm{fact}}$

$\Rightarrow$ $\mathrm{Sph}^{\mathrm{Spec}}_{G,XI}$ strictly factorizes

(a priori only lax)

<u>Construction of equivalence</u>

Following Gaitsgory - Lurie:

$$\mathrm{Sph}_{G,XI} \curvearrowright D\left(L^+G \backslash LG / LN^-, \psi\right)_{XI}$$

nondegenerate Whittaker

$$\downarrow$$

W_{XI} Whittaker vacuum

Naive Satake factorizes (Mirkovic - Vilonen, Raskin)

$$\mathrm{Rep}(\check{G})_{XI} \longrightarrow \mathrm{Sph}_{G,XI} \quad \text{monoidal fact. cat.}$$

<u>Th</u>. (Frenkel - Gaitsgory - Vilonen, Raskin)

$$\begin{tikzcd} \mathrm{Rep}(\check{G})_{XI} \arrow[r] \arrow[dr, "\sim"'] & \mathrm{Sph}_{G,XI} \arrow[d, "(-) * W_{XI}"] \\ & D\left(L^+G \backslash LG / LN^-, \psi\right)_{XI} \end{tikzcd}$$

$$\mathrm{Sph}_{G,X^I} \curvearrowright D(L^+G \backslash LG / LN^-, \psi)_{X^I} \overset{\text{FGV}}{\simeq} \mathrm{Rep}(\check{G})_{X^I}$$

$$\rightsquigarrow \mathrm{Sph}_{G,X^I} \longrightarrow \mathrm{End}\big(\mathrm{Rep}(\check{G})\big)_{X^I} \simeq \mathrm{Rep}(\check{G}\times\check{G})_{X^I}$$

monoidal fact. functor

$$\rightsquigarrow \mathrm{Sph}_{G,X^I} \simeq \delta_1\text{-mod}^{\text{fact}}\big(\mathrm{Sph}_{G,X^I}\big)$$

$$\longrightarrow \mathcal{O}_{\check{G}}\text{-mod}^{\text{fact}}\big(\mathrm{Rep}(\check{G}\times\check{G})\big)_{X^I}$$

$$\overset{\text{ren}}{\rightsquigarrow} \mathrm{Sph}_{G,X^I} \xrightarrow{\mathrm{Sat}_G} \mathrm{Sph}^{\mathrm{spec}}_{\check{G},X^I} \qquad \text{monoidal fact. functor}$$

Since Sat_G factorizes and preserves ULA objects, it suffices to check that it's an equiv. over $x \in X(k)$.

$$\begin{array}{ccc} \mathrm{Sat}_{G,x} : \mathrm{Sph}_{G,x} & \longrightarrow & \mathrm{Sph}^{\mathrm{spec}}_{\check{G},x} \\ \wr\wr & & \wr\wr \\ \mathrm{Sym}(\check{\mathfrak{g}}[-2])\text{-mod}\big(\mathrm{Rep}(\check{G})\big) & \simeq & \mathrm{IndCoh}\big(\mathcal{H}^{\mathrm{spec}}_{\check{G},x}\big) \end{array}$$

Koszul duality equiv.

It's far from clear that this square commutes.

Since $\mathrm{Sat}_{G,x}$ is $\mathrm{Rep}(\check{G})$-equivariant, it suffices to show that

$$\mathrm{Sym}(\check{\mathfrak{g}}[-2]) \simeq \underline{\mathrm{End}}_{\mathrm{Rep}(\check{G})}(\delta_1) \xrightarrow{\mathrm{Sat}_{G,x}} \underline{\mathrm{End}}_{\mathrm{Rep}(\check{G})}(\mathcal{O}_{\check{G}}) \simeq \mathrm{Sym}(\check{\mathfrak{g}}[-2])$$

is the identity.

~~NB: $\mathrm{Sym}(\check{\mathfrak{g}}[-2])^{\check{G}}$ generates $\mathrm{Sym}(\check{\mathfrak{g}}[-2])$ as $\check{G}$-equiv. algebra, so it suffices to check that~~

In fact, it suffices to check that

$$\mathrm{Sym}(\check{\mathfrak{g}}[-2])^{\check{G}} \simeq \mathrm{End}(\delta_1) \xrightarrow{\mathrm{Sat}_{G,x}} \mathrm{End}(\mathcal{O}_{\check{G}}) \simeq \mathrm{Sym}(\check{\mathfrak{g}}[-2])^{\check{G}}$$

Proceed by reduction to the case $G = T$.

That case follows from:

Th. (Beilinson)

$$\begin{tikzcd} D(LT)_{\times I} \arrow[r, "\sim"] & \mathrm{QCoh}(LS_{\check{T}}(\mathring{D}))_{\times I} \\ D(Gr_T)_{\times I} \arrow[u] \arrow[r, "\sim"] & \mathrm{QCoh}(LS_{\check{T}}(D))_{\times I} \arrow[u] \arrow[r, "\simeq"] & \mathrm{Rep}(\check{T})_{\times I} \end{tikzcd}$$

This is much stronger than derived Satake for T.

For the general case: use a certain $(\mathrm{Sph}_G, \mathrm{Sph}_T)$-bimodule.

$$\mathrm{Sph}_{G,B} := \left(D(L^+G \backslash LG)_{LN L^+T}\right)^{\mathrm{ren}}$$

Counterpart on the spectral side: there is a factorization algebra

$$\Upsilon(\check{n}, \mathcal{O}_{\check{G}}) \in \mathrm{Rep}(\check{G} \times \check{T}) \quad \text{s.t.}$$

$\Upsilon(\check{n}, \mathcal{O}_{\check{G}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G} \times \check{T}))$ is coherent, etc., and

$$\mathrm{Spec}^{\mathrm{Spec}}_{\check{G},\check{B}} := \left(\Upsilon(\check{n}, \mathcal{O}_{\check{G}})\text{-mod}^{\mathrm{fact}}(\mathrm{Rep}(\check{G} \times \check{T}))\right)^{\mathrm{ren}} \quad \text{satisfies}$$

$$\mathrm{Sph}^{\mathrm{Spec}}_{\check{G},\check{B},x} \simeq \mathrm{IndCoh}\left((\{0\} \times_{\check{\mathfrak{g}}} \check{n}) / \widehat{\check{B}_{\check{T}}}\right)$$

Th. (Raskin) There is a factorization functor

$$D(L^+G \backslash LG)_{LN\, L^+T} \longrightarrow \mathrm{Rep}(\check{G} \times \check{T})$$

$$\underset{\substack{\shortparallel \\ \text{vacuum}}}{\Delta^0} \longmapsto \Upsilon(\check{n}, \mathcal{O}_{\check{G}})$$

$\rightsquigarrow$ $\mathrm{Sat}_{G,B} : \mathrm{Sph}_{G,B} \longrightarrow \mathrm{Sph}^{\mathrm{Spec}}_{\check{G},\check{B}}$ "factorizable ABG functor"

Consider the following diagram:

$$\begin{tikzcd}
\mathrm{Sph}_{G,x} \arrow[r, "\mathrm{Sat}_{G,x}"] \arrow[d, "(-)*\Delta^0"'] & \mathrm{Sph}^{\mathrm{Spec}}_{\check{G},x} \arrow[d, "(-)*\Upsilon(\check{n},\mathcal{O}_{\check{G}})"] \\
\mathrm{Sph}_{G,B,x} \arrow[r] & \mathrm{Sph}^{\mathrm{Spec}}_{\check{G},\check{B},x} \\
\mathrm{Sph}_{T,x} \arrow[u, "\Delta^0*(-)"] \arrow[r, "\mathrm{Sat}_T", "\sim"'] & \mathrm{Sph}^{\mathrm{Spec}}_{\check{T},x} \arrow[u, "\Upsilon(\check{n},\mathcal{O}_{\check{G}})*(-)"']
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{Sym}(\check{\mathfrak{g}}[-2])^{\check{G}} \shortparallel \\
\mathrm{End}_{\mathrm{Sph}_G}(\delta_1) \arrow[r, "="] \arrow[d, "\text{Chevalley}"] & \mathrm{End}_{\mathrm{Sph}^{\mathrm{Spec}}_{\check{G}}}(\mathcal{O}_{\check{G}}) = \mathrm{Sym}(\check{\mathfrak{g}}[-2])^{\check{G}} \\
\mathrm{Sym}(\check{\mathfrak{t}}[-2]) \shortparallel = \mathrm{End}_{\mathrm{Sph}_{G,B}}(\Delta^0) \arrow[r] & \mathrm{End}_{\mathrm{Sph}_{\check{G},\check{B}}}(\Upsilon(\check{n},\mathcal{O}_{\check{G}})) \shortparallel \mathrm{Sym}(\check{\mathfrak{t}}[-2]) \\
\mathrm{End}_{\mathrm{Sph}_T}(\delta_1) \arrow[u] \arrow[r] \shortparallel & \mathrm{End}_{\mathrm{Sph}^{\mathrm{Spec}}_{\check{T}}}(\mathcal{O}_{\check{T}}) \arrow[u] \shortparallel \\
\mathrm{Sym}(\check{\mathfrak{t}}[-2]) \arrow[r, equal] & \mathrm{Sym}(\check{\mathfrak{t}}[-2])
\end{tikzcd}$$